



Cambrian College

School of Business and Information Technology, Creative Arts, Design, Music and Hospitality

Course Outline

Cambrian College would like to acknowledge that we are situated on the traditional lands of the Robinson Huron Treaty, and acknowledge our host, Atikameksheng Anishnawbek, as well as our ancestors of this land, and the Anishnaabe people.

Course Title	Programming for Analytics				
Course Code:	ANA1001	Credit Value:	4	Credit Hours:	56
Programs:	BAPG Business Analytics CAGC Crime Analytics HAGC Health Analytics				
Equivalencies		Prerequisites		Corequisites	

This course may be delivered in a variety of different formats: 100% in-class, 100% online (or a blend of both), videoconferencing, distributed learning or off-campus. Please confirm with your faculty member which format will be used for your section of this course.

General Education Course: ☐

Degree Breadth Course: ☐

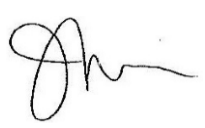
Eligible for PLAR: ☒

COURSE DESCRIPTION

In this course, students will explore the principles of programming and application design. Students will learn about fundamental programming tools, data types, arrays, control structures and fundamental algorithms. Students will solve various problems in code and move to breaking down larger problems into a series of smaller problems. Students will evaluate and create simple algorithms and be introduced to built-in structures. Students will analyze data and create visualizations. Students will also work with custom, programmer defined objects. Data persistence will be managed using File IO.

Date: July 21, 2021

Prepared By: Sidney Shapiro

Approved by: 

JILL FERGUSON
Chair, School Of Business And Information Technology,
Creative Arts, Design, Music And Hospitality

Effective: Fall 2021, Winter 2022, Spring 2022

RELATIONSHIP TO PROGRAM VOCATIONAL LEARNING OUTCOMES

PROGRAM LEVEL	
This course contributes to your program by allowing you to demonstrate the following vocational learning outcomes:	
Program(s)	Vocational Learning Outcomes
Business Analytics	College Standards 1. Collect, manipulate and mine data sets to meet organizational need. 2. Recommend different systems architectures and data storage technologies to support data analytics. 3. Design and apply data models that meet the needs of a specific operational/ business process. 4. Develop software applications to manipulate data sets, correlate information and produce reports. 5. Design and present data visualizations to communicate information to business stakeholders. 6. Apply data analytics, business intelligence tools and research to support evidence-based decision making. 7. Identify and assess data analytics business strategies and workflows to respond to new opportunities or provide project solutions.
Crime Analytics	College Standards 1. Collect and analyze a variety of data sets to identify information related to crime analysis. 2. Design and apply data models that meet the needs of a specific operational process. 3. Design and present data visualizations to communicate information to stakeholders.
Health Analytics	College Standards 1. Collect and analyze a variety of data sets to identify information related to organizational health needs. 2. Mine and manipulate data sets, correlate information and produce outputs using advanced software functions. 3. Design visualizations and communicate results of data analysis to decision makers, stakeholders and the public.

COURSE CURRICULUM**Topics/Concepts Covered in This Course**

- Computers and Computer Programming
- Modular programming in Python using Functions and Modules
- Statements, Variables, Decision and Looping Structures
- Data Structures and Object Oriented Programming
- Algorithms
- File IO

Topics/Concepts Covered in This Course

- Functions and classes
- Designing programs with user input
- Data visualization

COURSE LEVEL: Learning Outcomes and Objectives	
To earn credit for this course, you must reliably demonstrate your ability to:	
Learning Outcome	Objectives
1. Use Python to write a simple program.	1.1 Describe how computers store data. 1.2 Describe how a program works. 1.3 Run the Python interpreter. 1.4 Run the development interface software 1.5 Write and execute a basic script.
2. Create a script that reads input, performs a task and outputs a response.	2.1 Use comments in code. 2.2 Define and use strings and variables in code. 2.3 Read data input from the keyboard. 2.4 Write data output with the print Function. 2.5 Perform basic calculations within a script.
3. Write scripts that respond to varying conditions.	3.1 Identify good and bad if statements. 3.2 Identify good and bad if..else statements 3.3 Compare String values to determine equality. 3.4 List, describe and use logical operators in boolean expressions. 3.5 Describe and use boolean variables. 3.6 Code nested if...elif...else statements.
4. Write scripts that perform repetitions.	4.1 Code a condition controlled loop using a while statement. 4.2 Code a counter controlled loop using a for statement. 4.3 Describe what a sentinel value is in relation to loop structures.
5. Write modular programs using functions	5.1 Define and then call a function. 5.2 Design a program and use functions to break large problems into a series of smaller problems. 5.3 Differentiate between local variables, global variables and global constants. 5.4 Pass data as arguments to functions.
6. Write functions that return values.	6.1 Create a function that generates and returns a random number. 6.2 Write custom, value-returning functions.

Learning Outcome	Objectives
	6.3 Use pre-built functions in the Math module to process scripts. 6.4 Create your own custom modules to store like functions.
7. Process files and handle exceptions.	7.1 Read data from and write data to text files. 7.2 Use loops to process data stored in files. 7.3 Process records of data. 7.4 Handle anticipated problems, called exceptions. 7.5 Describe the process that occurs when an unanticipated problem occurs.
8. Use List and Tuple data structures.	8.1 Use Lists of data in scripts. 8.2 Searching and slicing lists of data. 8.3 Use built-in lists of data. 8.4 Copy Lists of data. 8.5 Process Lists of data.
9. Use String data within a script.	9.1 Search and slice string data. 9.2 Use built-in string manipulation functions.
10. Use Dictionary and Set data structures.	10.1 Differentiate between Lists and Dictionaries. 10.2 Use Dictionaries in scripts. 10.3 Serialize and deserialize object data.
11. Write programs using custom objects.	11.1 Write programs using procedural code. 11.2 Write programs using object oriented code. 11.3 Define and create custom classes. 11.4 Define and use instance variables.
12. Use external files	12.1 Describe loading data from external files 12.2 Describe the process of connecting to a database 12.3 Describe connecting to various Application Programming Interfaces (API)
13. Create data visualizations	13.1 Describe process to download and load data 13.2 Organize and analyze data 13.3 Use software to generate visualizations

Essential Employability Skills

Communication

- communicate clearly in written, spoken, and visual form that fulfills purpose/needs of audience.
- respond to written, spoken, or visual messages in a manner that ensures effective communication.

Numeracy

- execute mathematical operations accurately.

Critical Thinking and Problem Solving

- apply a systematic approach to solve problems.
- use a variety of thinking skills to anticipate and solve problems.

Information Management

- locate, select, organize, and document information using appropriate technology and info systems.
- analyze, evaluate, and apply relevant information from a variety of sources.

Interpersonal

- not applicable

Personal

- manage the use of time and other resources to complete projects.
- take responsibility for one's own actions, decisions, and consequences.

Delivery Method

- Classroom: Course is delivered through scheduled synchronous teaching that may be face-to-face and/or virtual.
- Online: Course is fully delivered through asynchronous teaching.
- Hybrid: Course combines scheduled synchronous and unscheduled asynchronous teaching.
- HyFlex: Course includes both synchronous and asynchronous learning and the student can move between both components seamlessly.

Learning Activities

- Lectures
- Class Discussions
- Labs
- Research
- Self-Directed Learning
- In-Class Exercises
- Case Studies
- eLearning Components

Resources Required

Books

Eric Matthes, *Python Crash Course, 2nd Edition*, 2nd Edition, No Starch Press
ISBN: 9781593279288

Evaluation Plan

Grading Scheme

A	80% - 100%
B	70% - 79%
C	60% - 69%
D	50% - 59%
F	0% - 49%

Evaluation Method	Value (%)
Assignments	35%
There are multiple code assignments in the course, please see your instructor for details	
Quizzes	30%
There are multiple quizzes in the course, please see your instructor for details	
Test/Exam	35%
There is a midterm and final project/exam in the course, please see your instructor for details	

ADDITIONAL INFORMATION

College

Modifications to the Course Evaluation Plan

Under exceptional circumstances, Cambrian reserves the right to alter the Course Evaluation Plan. Upon approval from the Dean, faculty members will notify students of any modification to the Course Evaluation Plan and post to the course Learning Management System site (Moodle).

Academic Policies

Students must adhere to all academic policies. These are available on myCambrian as well as the College website at <https://cambriancollege.ca/about/official-documents-and-policies/academic-policies/>.

Accessibility Services

Cambrian is committed to creating an inclusive learning environment where we support our students' success. Students who are registered with the Glen Crombie Centre can access accessibility and/or counselling services by:

1. Registering and scheduling an appointment through Clockwork in the Student Portal in myCambrian.
2. Contacting 705-566-8101 ext. 7420 or 7311, or
3. E-mailing accessibilityservices@cambriancollege.ca or Counselling@cambriancollege.ca

Attendance

Regular attendance is strongly encouraged as it contributes to student success. Some courses have specific lab/participation requirements. Students are expected to be aware of, and adhere, to these requirements.

Audio/Visual Capture

Sounds and images from this class, and contributions made by a participant, virtually or in-person, are recorded under the authority of the Ontario Colleges of Applied Arts and Technology Act, 2002. The main purpose for these recordings is to allow students enrolled in this course, whether they attend the particular class in person, virtually, or at all, to review class content and activity. The use of class recordings is for personal use only and shall not be shared or transferred.

These recordings may also be reviewed by faculty members to prepare future classes, to evaluate students, to collaborate in program/course development/review, or to provide feedback to students or other faculty. Any questions about this collection may be addressed to your respective Dean.

Copyright

Students may not submit work or partake in activities that infringe on the Canadian Copyright Act, or the related Cambrian College Fair Dealing Guidelines and Copying Guidelines. The guidelines may be viewed at: <http://cambriancollege.libguides.com/copyrightinfo> or by contacting the Library for questions related to copyright: library@cambriancollege.ca

Wabnode Centre for Indigenous Services

Students are encouraged to come to Wabnode to meet our Elders and to access personal counselling and connect with academic tutoring. Our team is also able to provide students with information related to funding and scholarship and bursary opportunities for Indigenous students. We encourage students to come to Wabnode and find out about opportunities to share in cultural, sports and community activities. Students can also contact Wabnode at wabnode@cambriancollege.ca

Out-of-Class Assistance

The faculty will inform students of their Out-of-Class availability through the Learning Management System site (Moodle).

Prior Learning Assessment and Recognition (PLAR)

Students wishing to have work or life experience considered through Prior Learning Assessment and Recognition should contact pathways@cambriancollege.ca

Testing Practices at Cambrian

Many courses include major tests and/or final exams. The practice at Cambrian requires that these types of test situations involve proctoring to ensure academic integrity. Online tests/exams may employ a proctoring service to enable you to take your exam from a location of your choosing within a period specified by your instructor. When you are taking an online test/exam, the proctoring service may capture your video, screen, audio and web surfing particulars to protect academic integrity. Cambrian College collects, uses, discloses and retains personal information in compliance with the Freedom of Information and Protection of Privacy Act (FIPPA). Your personal information is being collected under the authority of the Ontario Colleges of Applied Arts and Technology Act S.O. 2002, c.8, Sched. F. This information will be used for the purpose of administering a test/exam through an online proctoring service acting as an authorized agent of the College. Please refer to Cambrian's Confidentiality of Student Records Policy for more details. If you have any questions regarding the collection of your personal information, please contact Vice President Academic, Cambrian College, 1400 Barrydowne Rd., Sudbury ON P3A 3V8, 1-705-566-8101.

Tests and Evaluations

Students are required to write tests and/or complete evaluations as scheduled. Exceptions may be made in the event of an emergency or a sanctioned event (e.g., varsity sports, field trip, religious observances, etc.). In non-emergency situations, students are expected to contact the faculty member in the event that they can not be present for a test/evaluation to explain the reason for their absence. At the discretion of the faculty member, an alternate time for the test/evaluation may be allowed.

Transfer Credit

Students wishing to have courses from other programs or institutions assessed for equivalency and/or transfer credit should contact transfercredits@cambriancollege.ca.

Equity, Diversity and Inclusivity

Cambrian is committed to building and preserving an equitable, diverse, and inclusive learning community where students, faculty, and staff may achieve their full potential in an environment characterized by equality of respect and opportunity. All students and employees have the right to live and work in an environment that is free from discrimination and harassment. Therefore, Cambrian College will not tolerate any form of discrimination or harassment in its employment, education, accommodation, or business dealings. For more information, please visit: <https://cambriancollege.ca/about/official-documents-and-policies/equity-human-rights-and-accessibility/>